Factors & Multiples

The Magic of Sharing and Grouping

Target Grade: 3 | Session 1

Session Overview

In this session, we will explore the wonderful world of numbers. We will learn how to break numbers down into their "builders" (Factors) and how to grow them into "families" (Multiples) using fun Indian examples like Ladoos, Auto-Rickshaws, and Rangolis!

Learning Objectives

By the end of this lesson, our young mathematicians will be able to:

- Understand **Factors** as numbers that multiply together to give a product.
- Identify **Multiples** through skip-counting and multiplication.
- Apply these concepts to solve real-life sharing problems.
- Use visual tools like "Arrays" and "Ladoo Boxes" to represent mathematical relationships.

TEACHER'S TIP

Encourage students to use physical objects like rajma beans or marbles during the "Array" section of the lesson to make the concept of factors tangible.

The Great Indian Picnic

Imagine it is a sunny Saturday afternoon. You and your friends—Ananya, Ishaan, Kabir, and Zara—are sitting on a colorful mat at the local park.

[Visual: Illustration of children sharing snacks on a picnic mat]

The Snack Challenge

You have brought a box of **12 delicious Ladoos**. Everyone is hungry!

The Question: How can we share these 12 Ladoos equally so that no one is left out and no Ladoo is left over?

Think about the different ways you can group them:

- Could 1 person eat all 12? (1 group of 12)
- Could 2 people have 6 each? (2 groups of 6)
- Could 3 people have 4 each? (3 groups of 4)

All these numbers we used—1, 2, 3, 4, 6, and 12—are the **Factors** of 12!

What Are Factors?

Factors are the "Builders" of numbers. They are the small numbers that come together to create a bigger number.

> **Definition:** Factors are numbers that you multiply together to get another number (the product).

Building 12

Look at these different "building" combinations:

$$1 \times 12 = 12$$

$$2 \times 6 = 12$$

$$3 \times 4 = 12$$

In each case, the two numbers on the left are the **factors**, and 12 is the **product**.

THINK & ANSWER

If you have 6 Gulab Jamuns, can you find all the factors? (Hint: In what ways can you arrange them equally?)

The Array Method

A great way to "see" factors is by using **Arrays**. An array is just a way of arranging objects in equal rows and columns.

Using Rajma Beans

Let's use Rajma beans to find the factors of 8.

Option A: 1 row of 8 beans
$1 \times 8 = 8$

Option B: 2 rows of 4 beans
$2 \times 4 = 8$

Option C: 4 rows of 2 beans
$4 \times 2 = 8$

Can we make 3 rows of beans? Let's try... 3, 6... the next is 9. We can't make exactly 8! This means **3 is NOT a factor** of 8.

Interactive Practice: Factor Finders

Let's practice finding the "builders" for some other common items we see every day.

Factors of 10 (The Ladoo Boxes)

Grouping Style	Math Equation	Factors Found
1 box of 10	1×10	1, 10
2 boxes of 5	2×5	2, 5

Factors of 10: **1, 2, 5, 10**

Factors of 15 (Marigold Garlands)

We are making garlands for a festival. We have 15 Marigold (Genda) flowers.

- 1 garland of 15 flowers: 1×15
- 3 garlands of 5 flowers: 3×5

Factors of 15: **1, 3, 5, 15**

What Are Multiples?

If factors are the builders, **Multiples** are the "Family." They are what you get when a number grows by multiplying it with 1, 2, 3, and so on.

> **Definition:** A multiple is the result (product) of multiplying a number by any whole number.

The Auto-Rickshaw Example

Think of the wheels on an Auto-Rickshaw. One auto has 3 wheels.

- 1 Auto: $3 \times 1 = 3$ wheels
- 2 Autos: $3 \times 2 = 6$ wheels
- 3 Autos: $3 \times 3 = 9$ wheels
- 4 Autos: $3 \times 4 = 12$ wheels

The numbers **3, 6, 9, 12, 15...** are all multiples of 3!

The Skip-Counting Game

Multiples are exactly like **skip-counting**. Let's play a game using Indian currency notes (pretend ones!).

COUNTING ₹5 NOTES

If you keep adding ₹5 notes to your piggy bank, how much money do you have?

1. One note: ₹5
2. Two notes: ₹10
3. Three notes: ₹15
4. Four notes: ₹20

You just listed the first four multiples of 5: **5, 10, 15, 20**.

Quick Challenge

Can you find the first five multiples of 2 by skip-counting?
Answer: 2, 4, 6, 8, 10

Factors vs. Multiples: The Difference

It is easy to get these two mixed up! Let's look at the "Rule of Size" to help us remember.

Feature	Factors	Multiples
Definition	Numbers that build the product.	The "Family" of products.
Size	Usually **smaller** or equal to the number.	Usually **larger** or equal to the number.
Amount	There is a **fixed** number of them.	They go on **forever**! (Infinite)

Example: The Number 6

Factors: 1, 2, 3, 6 (They stop at 6!)

Multiples: 6, 12, 18, 24, 30... (They keep going!)

Daily Life Challenge

Math is all around us in India! Your mission is to find factors and multiples in your own home today.

Challenge Checklist

- **The Kitchen Challenge:** Find a pack of biscuits. How many are inside? Can you find the factors of that number?
- **The Street Challenge:** Count the wheels on the cars parked on your street. Is the total number a multiple of 4?
- **The Rangoli Challenge:** Look at a Rangoli pattern. Does it have repeating shapes in groups of 3 or 4? That's an array!

Lesson Recap

Congratulations! You have mastered the basics of Factors and Multiples. Let's remember the big ideas:

1. **Factors** are the numbers you multiply to get a product. Think of them as "Building Blocks."
2. **Multiples** are the results of multiplying a number. Think of them as "Skip-Counting."
3. You can use **Arrays** (rows and columns) to find factors visually.
4. **Indian Context:** Whether it's counting Ladoos or Auto-Rickshaw wheels, math helps us understand our world better!

Great Job, Mathematician!

Toh chalo, see you in the next session!

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